Where Does the Gain Come From? A Controlled Study of Training-Free RoPE Extrapolation to 4K in Diffusion Transformers

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Abstract

Training-free methods let a diffusion transformer trained near one megapixel generate 4096×4096 images by editing its rotary position embedding (RoPE): they raise the frequency base, add a global attention temperature and, in the recent SEGA, rescale each rotary band by a gain read from the latent spectrum. We ask which edit produces the improvement. On FLUX.1-Krea-dev, with a dev/held-out split fixed in advance and paired bootstrap intervals over 187 held-out prompts, SEGA's base and global temperature alone recover 104% [97, 113] of its ImageReward gain over an NTK base and tie the released method on all three reward models and on crop MUSIQ; the per-band allocation adds only 2.09 full-frame MUSIQ points. Four per-band laws that we derive from efficient coding and contrast constancy do no better: the two evaluated held out tie or lose to SEGA on every reward and image-quality metric, and trajectory diagnostics show why. Results also depend on the checkpoint: FLUX.1-dev breaks into a halftone lattice at 4096^2 even under the official DyPE code, while FLUX.1-Krea-dev stays clean. SEGA's printed equations differ from its released code at every stage. Finally, 224-pixel reward models cannot see the top 4 octaves of a 4K image, and the default flow-shift extrapolation collapses the noise schedule.



Figure 1: SEGA's 4K gain over the NTK base comes mostly from its RoPE base and global temperature; the per-band allocation adds little. (a) Held-out row 28, seed 0: full frame of SEGA (released) and native 384-pixel crops at identical coordinates; each column adds a component of SEGA, and compositions differ between arms. (b) Share of the gain of SEGA (released) over NTK that SEGA base + temp. recovers, with 95% paired bootstrap intervals over 187 held-out prompts; the dark segment is the allocation's part.

1 Introduction

Diffusion transformers trained at about one megapixel are routinely asked for 4096×4096 images, 16 times more tokens than any training image. Training-free methods make this work by editing the rotary position embedding (RoPE) [47] at inference: they rescale its frequencies [7, 10, 41], schedule the rescaling with the noise level [27], sharpen attention with a global logit temperature [28, 37] and,

most recently, scale each rotary band by a gain read from the spectrum of the current latent [45]. Each method is compared as a package with the previous one, and the improvement is credited to its newest component.

We ask where the improvement comes from. We take SEGA [45], a strong recent method with released code, on the checkpoint its authors use (FLUX.1-Krea-dev), and separate its three components: a strengthened NTK frequency base, a global temperature and a content-adaptive per-band allocation of query and key gains. All arms share prompts, seeds, schedule and sampler; the dev/held-out split was fixed before any held-out score was seen; every main comparison is paired over 187 held-out prompts with bootstrap intervals, and a seed-1 replication covers 55 of them. We then test whether per-band laws with a principled basis beat SEGA's heuristic: from efficient coding and contrast constancy we derive four controllers, select among them on the dev set under a written gate, and evaluate the survivors held out. Our findings, in order of strength:

(i) **Base and temperature carry most of the gain** (Figure 1). Of SEGA's gain over the NTK base, its base and global temperature alone carry 104% on ImageReward, 103% on PickScore and 99% on crop MUSIQ. The per-band allocation adds a small but real full-frame sharpness gain (MUSIQ +2.09 [+0.96, +3.25]) and nothing measurable on the reward models (Section 5.1).

(ii) **Principled per-band laws do not beat the heuristic.** The two controllers that reached the full held-out set tie or lose to SEGA (released) on every reward and image-quality metric; the closer one, SEGA + TM, ties it on ImageReward and full-frame MUSIQ and loses on PickScore (−0.08) and crop MUSIQ (−1.13). Trajectory diagnostics trace the failure to a gate that opens late, a target that trades sharpness for spectral match, an allocation that is mostly a fixed tilt and a weak gain actuator (Section 5.2).

(iii) **Results depend on the checkpoint.** FLUX.1-dev renders a halftone lattice at 4096^2 even under the official DyPE code, while FLUX.1-Krea-dev is clean (Section 5.3).

(iv) **SEGA's paper and code differ.** The released code changes the sensor input, the gain law, its bounds and the frequency base. In a descriptive eight-prompt pilot the printed equations score worse than the released code on every metric, and their static tilt destroys images (Section 5.4).

(v) **4K evaluation has traps.** 224-pixel reward models cannot see the top 4 octaves of a 4K image; the default flow shift extrapolates to $\mu = 11.55$, so every model call sees almost pure noise; and forced FlashAttention-2 takes 1.54 times as long as cuDNN attention on a GH200 (Section 5.5).

We do not propose a new state-of-the-art method. We contribute the controlled decomposition, a negative result with its mechanisms, and a protocol that makes such claims checkable. Code, per-image scores and the scripts that generate every table and number in the paper are released.

2 Background

RoPE in FLUX. FLUX [5] packs the autoencoder latent into 2×2 patches, so one token covers 16×16 pixels and a 4096^2 image has a 256^2 token grid (64^2 at 1024^2). Each 128-dimensional attention head has rotary blocks of 16, 56 and 56 dimensions for an index axis and the two spatial axes. Pair j of spatial axis a rotates at $\omega_{a,j} = \theta^{-2j/D}$ with $\theta = 10^4$ and $D = 56$, so wavelengths $2\pi/\omega_{a,j}$ run from about 6 to about 45,000 tokens over the 28 pairs.

The logit as a Fourier series over the offset. Write the query and key coefficients of pair j on axis a as complex numbers $\tilde{q}_{a,j}$ and $\tilde{k}_{a,j}$, and let a method multiply both by a gain $m_{a,j}$ after the rotation, with which it commutes. The logit between a query at position u and a key at position v is then

$$\ell_{uv} = \frac{1}{\sqrt{d_k}} \sum_{a,j} g_{a,j} \left|\tilde{q}_{a,j}\right| \left|\tilde{k}_{a,j}\right| \cos\!\left(\omega'_{a,j} \Delta^a_{uv} + \phi_{a,j}\right), \qquad g_{a,j} = m^2_{a,j}, \tag{1}$$

a Fourier series in the offset $\Delta^a_{uv} = p^a_u - p^a_v$ with one term per rotary pair, plus a non-rotating index-axis part. Content sets the magnitudes and phases $\phi_{a,j}$; extrapolation changes only the frequencies $\omega'_{a,j}$ and the gains $g_{a,j}$. A gain shared by all pairs, $g \equiv \tau$, is an inverse temperature; a j-dependent gain reweights the bands of the kernel.

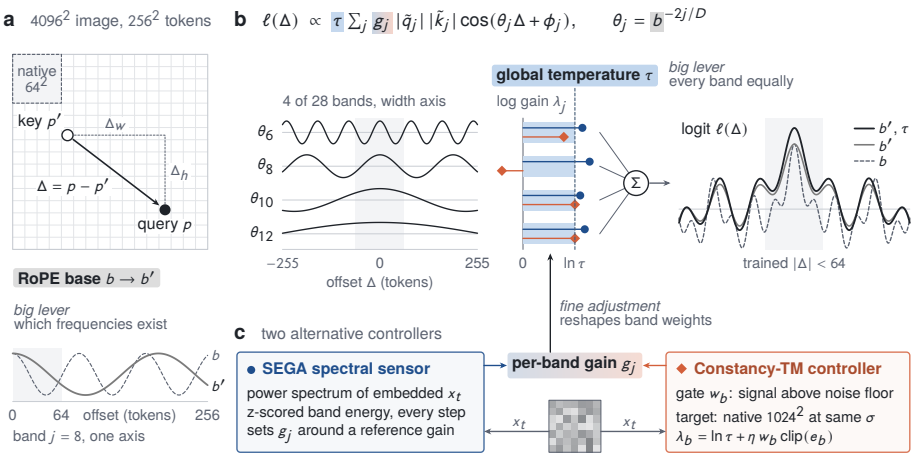


Figure 2: The RoPE base and one global temperature are the big levers on the attention logit; per-band gains fine-tune band weights. (a) Query, key and offset on the token grid of a 4096^2 image; the inset shows band 8 under the native base b (dashed) and SEGA's released base b'. (b) Four of the 28 width-axis bands with their log gains (sky: $\ln \tau$; circles: SEGA (released); diamonds: Constancy-TM) from one held-out sample at step 14; the curves on the right sum only these four bands. (c) Each controller reads x_t at every step and sets g_j.

Table 1: Arms and the knobs they turn. Temperature is the logit gain at 4096^2; t/x: varies with the noise level / with the image. Held-out arms ran on rows 8–194, the pilot and dev arms only on the eight dev prompts.

Arm	Frequency base	Temperature	Per-band gain	t/x	Evaluated
NTK	$\theta\, s^{2D/(D-2)}$	none	none	no / no	held-out
DyPE	PI/NTK ramps in σ	1.30 (spatial)	none	yes / no	held-out
SEGA (released)	$\theta\, s^{2D/(D-2)}/(1{+}0.1 \ln s)$	$s^{0.16} = 1.248$	sensor, ≥ 1	yes / yes	held-out
SEGA (paper eqs.)	$\theta\, s^{2D/(D-2)}$	$s^{0.16}$	sensor, unbounded	yes / yes	pilot
SEGA base + temp.	as SEGA (released)	$s^{0.16}$ on all dims	none	no / no	held-out
Constancy	as NTK	$\tau_N = 1.333$	gated, late target	yes / yes	partial (Appendix D)
Constancy-TM	as SEGA (released)	$s^{0.16}$	gated, time-matched	yes / yes	held-out
Alias-T	as SEGA (released)	none	alias count, static	no / no	dev
SEGA + TM	as SEGA (released)	$s^{0.16}$	SEGA × time-matched	yes / yes	held-out

Three knobs. We sort methods by the knobs they turn (Figure 2). *Frequency base*: at 4096^2 the offsets reach $s = 4$ times the native range, so slow pairs see phases never seen in training. Position interpolation divides every frequency by s [10]; NTK-aware scaling raises the base to $\theta' = \theta s^{D/(D-2)}$, which leaves the fastest pair alone and divides the slowest by s [7]; YaRN interpolates only the slow pairs [41]. *Global temperature*: with N keys instead of N_{train}, softmax entropy grows roughly with $\log N$, and multiplying logits by $\tau_N = \log N / \log N_{\text{train}}$ offsets it [28]; at 4096^2, $\tau_N = 1.333$. *Per-band gain*: a j-dependent $g_{a,j}$, possibly varying with the noise level and with the image.

The flow schedule at 4K. FLUX samples a rectified flow [34, 35] with a time shift $\sigma' = e^{\mu}/\left(e^{\mu} + 1/\sigma - 1\right)$ whose μ grows linearly with the token count [16]. The diffusers default [49] extrapolates that line to $\mu = 11.55$ at 4096^2. The last evaluated noise level is then $\sigma = 0.99974$: every model call sees almost pure noise, and the final Euler step jumps from there to the image. The official DyPE and SEGA pipelines cap μ at 1.15 (last $\sigma = 0.10472$), as do all our arms. Comparisons that mix the two schedules are not comparisons of position embeddings.

3 Methods under study

All arms share the sampler and differ only in ω' and g of Equation (1) (Table 1; exact formulas and code revisions in Appendix E).

Frequency baselines. NTK is the strengthened NTK base that SEGA builds on, $\theta' = \theta s^{2D/(D-2)}$, which divides the slowest pair's frequency by s^2. With no temperature and no allocation, it is the floor of the decomposition. DyPE [27], ported from the official code, blends position interpolation, NTK and native frequencies with two ramps in σ^2, so fine pairs return to their native frequencies late

Table 2: SEGA's printed equations and released code differ at every stage. Condensed from the released code at the inspected revision and our reading of the paper; s is the scale per side, $D = 56$.

Stage	Printed equations	Released code
Sensor input	channel mean of the latent	channel mean of the projected input features
Pair-to-bin map	physical frequency, clipped	rank of log period
Allocation	$m_{\text{ref}}(1 - \varsigma u)$, $u = \tanh z - \overline{\tanh z}$	$\max\big(1,\ m_{\text{ref}}(1 - 0.15\,\varsigma c)\big)$, $c = \tanh 1.5z - \overline{\tanh 1.5z}$
Bounds	none	gains clamped at 1
Index axis	m_{ref}	1
Frequency base	$\theta s^{2D/(D-2)}$ per axis	$\theta s^{2D/(D-2)}/(1 + 0.1 \ln s)$, longest side

in denoising; its cos/sin amplitude $1 + 0.1 \ln s$ acts as a logit gain of 1.30 on the spatial dimensions.

SEGA: paper equations and released code. SEGA [45] reads the spectrum of its input at every model call and sets per-pair gains around a reference gain $m_{\text{ref}} = s^{0.08}$. In the printed equations, the energy E_a of the channel-mean latent at pair j's frequency bin is standardized across pairs, $z_{a,j}$, and

$$m_{a,j} = m_{\text{ref}}\big(1 - \varsigma\, u_{a,j}\big), \qquad u_{a,j} = \tanh z_{a,j} - \text{mean}_i \tanh z_{a,i}, \qquad \varsigma = 1 - \text{SF}(E_{\text{iso}})^{1.5}, \tag{2}$$

where SF is the flatness of the radial spectrum. Low-energy pairs, usually the fast ones, thus gain attention weight in proportion to how far the spectrum is from flat. The released code differs from Equation (2) in every stage (Table 2). We call its parity-tested port SEGA (released) and our implementation of the printed equations SEGA (paper eqs.); static variants use the mean allocation over pilot prompts and steps.

SEGA base + temp.: the decomposition control. SEGA base + temp. keeps the frequency base of SEGA (released) and applies its reference logit gain $s^{0.16}$ uniformly to every dimension, with no allocation. Relative to SEGA (released) it removes the allocation's shape, the small lift in mean gain that the clamp at 1 produces, and the difference on the index axis (gain 1 in the released code).

3.1 Per-band laws from vision science

SEGA's gain law is a heuristic; we asked whether laws from biological vision do better. Perceived contrast in each spatial-frequency band stays nearly constant above threshold across viewing distances, and so across sampling rates [20]. Efficient coding explains this by weighting each band by its visibility against noise, the Wiener factor $S/(S+N)$ [1, 2, 48], and by renormalizing band contrast above threshold toward natural-image statistics [18, 39, 51]. A diffusion sampler knows what the retina does not: the exact noise power in every band at every step. Each law sets $\ln g_{a,j} = \ln \tau + \Delta_{a,j}$, a temperature plus a bounded per-band correction.

Constancy. For half-octave bands b, a gate compares the current input's band energy with the known noise floor, $\rho_{a,b} = \max\big(E_a^x[b]/(\sigma^2 C) - 1, 0\big)$ and $w_{a,b} = \rho_{a,b}/(1 + \rho_{a,b})$, and an error compares the previous clean estimate's band energy $\hat{E}_a[b]$ with a native-resolution target P^{nat}, the mean log spectrum of the model's own 1024^2 samples on 16 calibration prompts outside the evaluation set:

$$e_{a,b} = \tfrac{1}{2}\big(P_a^{\text{nat}}(f_b) + c_a - \ln \hat{E}_a[b]\big), \qquad \delta_{a,b} = \eta\, w_{a,b}\, \text{clip}(e_{a,b}, -1, 1), \tag{3}$$

where c_a anchors the target level on visible bands of 16 to 64 token wavelengths. The pair correction $\Delta_{a,j}$ interpolates δ in $\log f$, with $\tau = \tau_N$ and $\eta = 1$. A natural spectrum gives $e = 0$ and reduces the law to the global temperature. The target is the native spectrum of the final, nearly clean steps.

Constancy-TM. The same law with a time-matched target: $P^{\text{nat}}(f, \sigma)$ is the clean estimate's native spectrum at the same noise level, tabulated over all 28 native steps, so the anchor no longer needs the gate. Built on the base and temperature of SEGA (released), it isolates the allocation, with SEGA base + temp. as its control.

Alias-T and SEGA + TM. Alias-T applies the log-N temperature pair by pair, taking as a pair's key pool the number of keys it can resolve on the grid (Appendix E); Alias-T-s shifts all its log gains by one constant to match SEGA's reference gain in geometric mean. SEGA + TM multiplies the allocation of SEGA (released) by the time-matched correction, $\ln g = \ln g^{\text{SEGA}} + \Delta$, with $\eta \in \{0.5, 1\}$ on the dev set and $\eta = 1$ held out.

4 Experimental protocol

Model and sampler. All results use the FLUX.1-Krea-dev transformer [6], the checkpoint loaded by the official DyPE and SEGA scripts, with the FLUX.1-dev text encoders and autoencoder [5] (Appendix E). A 4096^2 image has 65,536 tokens, 16 times the 4,096 of a native 1024^2 image ($s = 4$ per side). Every arm uses 28 Euler flow-matching steps, distilled guidance 4.5, bfloat16, batch size one, the same per-prompt seeds and the capped flow shift $\mu = 1.15$ (Section 2).

Prompts and the dev/held-out split. Of the 195 evaluation captions of Aesthetic-4K [55], rows 0–7 form a development set, used for every choice of variant and hyperparameter, and rows 8–194 the held-out set ($n = 187$) on which all reported comparisons are made. We fixed this split after an eight-prompt pilot and before inspecting any held-out score. From the second campaign on, a written gate decided which variants could leave the dev set: against SEGA (released) on the eight dev prompts, a variant needed higher ImageReward and full-frame MUSIQ and a CLIP drop of at most 0.3, and at most two variants could advance per campaign. Dev numbers are selection evidence only (Appendix D.2). Seed 0 is used throughout; a seed-1 replication covers held-out rows 8–62 ($n = 55$) for SEGA (released), SEGA base + temp. and Constancy-TM.

Metrics. *Alignment and preference*: CLIP score [24, 43], the scaled cosine $100 \max(\cos, 0)$ of ViT-L/14 embeddings, PickScore [32] and ImageReward [53]. These models resize their input to 224 pixels, so we read them as measures of layout and alignment (Section 5.5). *No-reference quality*: MUSIQ [30] and CLIP-IQA [50], on the full 4096^2 frame ("full-frame") and averaged over four fixed 1024^2 crops shared by all methods ("crop"). *Spectral fidelity and repetition*: the root-mean-square and log-log slope errors of luminance power in 11 octave bands from 2^{-12} to 2^{-1} cycles per pixel, against the mean profile of the 195 Aesthetic-4K reference images, and a tiling index, the autocorrelation at a 1024-pixel lag minus that at 1536 pixels. FID and KID [4, 25] are too noisy at our sample sizes to rank methods (Appendix B). No metric here measures 4K quality on its own, and we ran no human study.

Statistics. Each comparison is a paired difference against SEGA (released) over the same prompts and seed, with a 95% percentile bootstrap interval over 2,000 prompt resamples. A difference counts as better or worse only when its interval excludes zero; an interval that covers zero is a tie in our wording, not equivalence. Intervals are per metric and not adjusted for multiple comparisons.

5 Results

5.1 Decomposing SEGA (released): base and temperature do most of the work

Table 3 and Figure 3 compare every held-out arm with SEGA (released). Two rows carry the decomposition. NTK keeps SEGA's family of frequency base and drops both the temperature and the allocation; it is far worse on every reward and image-quality metric (ImageReward −0.594, full-frame MUSIQ −15.74, crop MUSIQ −18.70). SEGA base + temp. keeps SEGA's released base and global temperature and drops only the allocation. It ties SEGA (released) on ImageReward (+0.026 [−0.016, +0.070]), PickScore, CLIP and crop MUSIQ, and falls behind on full-frame MUSIQ (−2.09 [−3.25, −0.96]), CLIP-IQA and the spectral error.

Figure 1b and Table B.1 (Appendix B) turn these rows into shares. Of SEGA's gain over NTK, base and temperature carry 104% of the ImageReward gain (interval [97, 113]), 103% of PickScore, 101% of CLIP and 99% of crop MUSIQ. The allocation's own contribution, SEGA (released) minus SEGA base + temp., is +2.09 [+0.96, +3.25] on full-frame MUSIQ and +0.009 [+0.001, +0.018] on CLIP-IQA; its intervals cover zero on all three reward models and on crop MUSIQ. SEGA's per-band allocation is therefore real but narrow: it sharpens the full frame as MUSIQ sees it and leaves alignment, preference and crop-level quality where the base and global temperature put them.

DyPE makes a different trade. Against SEGA (released) it is worse on all three reward models (ImageReward −0.164 [−0.258, −0.087], PickScore −0.49) and better on crop MUSIQ (+1.97 [+1.16, +2.80]) and CLIP-IQA (+0.029). No single metric orders the methods, so we report all of them.

Table 3: Base and temperature carry most of SEGA's gain, and no per-band law beats it. Held-out paired differences against SEGA (released) (rows 8–194, seed 0, $n = 187$); the first row gives the absolute means of SEGA (released). Each cell shows the mean difference with its 95% bootstrap interval below; ▲/▼: the interval excludes zero in the better or worse direction; uncoloured cells are ties. Table B.3 has all ten metrics.

Method	Image-Reward↑	Pick-Score↑	CLIP score↑	MUSIQ full↑	MUSIQ crop↑	CLIP-IQA full↑
SEGA (released)	1.190	23.05	29.14	45.23	58.33	0.603
Decomposition of SEGA (released)						
NTK	−0.594 ▼ [−0.701, −0.494]	−1.02 ▼ [−1.16, −0.89]	−1.99 ▼ [−2.43, −1.55]	−15.74 ▼ [−17.50, −14.00]	−18.70 ▼ [−20.68, −16.86]	−0.046 ▼ [−0.062, −0.031]
SEGA base + temp.	+0.026 [−0.016, +0.070]	+0.03 [−0.03, +0.09]	+0.02 [−0.19, +0.23]	−2.09 ▼ [−3.25, −0.96]	−0.11 [−0.71, +0.51]	−0.009 ▼ [−0.018, −0.001]
Other baseline						
DyPE	−0.164 ▼ [−0.258, −0.087]	−0.49 ▼ [−0.63, −0.36]	−0.37 ▼ [−0.72, −0.01]	−1.88 ▼ [−3.26, −0.56]	+1.97 ▲ [+1.16, +2.80]	+0.029 ▲ [+0.020, +0.038]
Principled per-band laws (ours)						
Constancy-TM	+0.018 [−0.023, +0.059]	−0.05 [−0.11, +0.02]	−0.27 ▼ [−0.49, −0.04]	−3.04 ▼ [−4.28, −1.81]	−2.19 ▼ [−2.81, −1.57]	−0.025 ▼ [−0.034, −0.017]
SEGA + TM	+0.009 [−0.008, +0.027]	−0.08 ▼ [−0.10, −0.05]	−0.09 ▼ [−0.20, −0.00]	−0.21 [−0.71, +0.32]	−1.13 ▼ [−1.39, −0.90]	+0.003 [−0.000, +0.006]

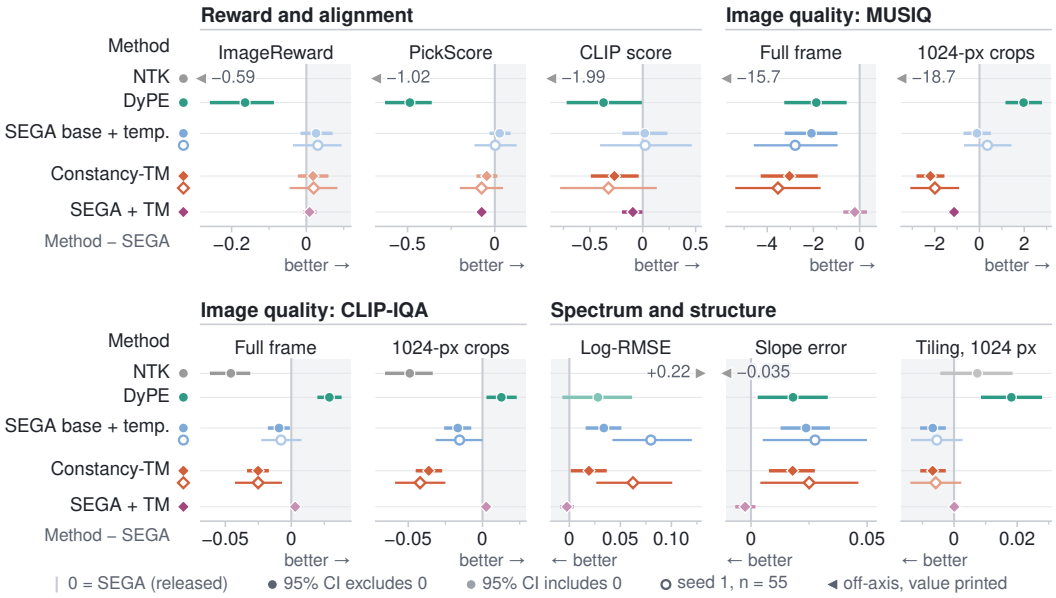


Figure 3: No per-band law improves on SEGA (released) in any reward or image-quality metric. Paired held-out differences from SEGA (released) with 95% bootstrap intervals (rows 8–194, seed 0, $n = 187$). Hollow markers: seed-1 replication on rows 8–62 ($n = 55$) for SEGA base + temp. and Constancy-TM. Shaded half: better direction. Isolated gains, such as crop MUSIQ and CLIP-IQA for DyPE, come with losses elsewhere.

Replication at seed 1. On the 55 seed-1 prompts, SEGA base + temp. again ties SEGA (released) on ImageReward, PickScore and CLIP and loses on full-frame MUSIQ (−2.79 [−4.58, −0.95]). 17 of the 20 tie/better/worse verdicts for SEGA base + temp. and Constancy-TM agree with seed 0 on the same prompts (Table C.1).

Qualitative. Figure A.1 (Appendix A) shows three held-out prompts: NTK loses fine detail, and the other arms differ mainly in native-resolution crops, which 224-pixel scorers never see.

5.2 Principled per-band laws do not beat SEGA (released)

Outcome. Neither law evaluated on the full held-out set improves on SEGA (released) in any reward or image-quality metric (Table 3, bottom block). Constancy-TM ties it on ImageReward and PickScore and is worse on CLIP (−0.27), full-frame MUSIQ (−3.04), crop MUSIQ (−2.19), CLIP-IQA and the spectral error; its one gain is a lower tiling index (−0.0068). It also loses to

SEGA base + temp., which shares its base and temperature and has no allocation: crop MUSIQ −2.08 [−2.43, −1.75] and full-frame MUSIQ −0.95, against small gains in both spectral errors (−0.015 and −0.006). SEGA + TM at $\eta = 1$, the only variant to pass the campaign-2 dev gate, ties SEGA (released) on ImageReward (+0.009 [−0.008, +0.027]) and full-frame MUSIQ (−0.21 [−0.71, +0.32]) and loses on PickScore (−0.08 [−0.10, −0.05]) and crop MUSIQ (−1.13 [−1.39, −0.90]); its CLIP interval, [−0.20, −0.00], ends just below zero. Alias-T and Alias-T-s failed the gate on ImageReward and were never run held out (Table D.1); the original Constancy has only a partial held-out run (Appendix D). Four diagnostics, on dev prompts and on the logged gains of held-out runs, explain the outcome (Figure F.1 in Appendix F); they are explanations, not further comparisons.

The visibility gate opens too late. Constancy's correction is exactly zero until a band with a wavelength of 16 to 64 tokens holds more than twice the noise power ($w > \frac{1}{2}$). On the eight dev prompts it first becomes nonzero at $\sigma = 0.60$ and is nonzero on all of them only by $\sigma = 0.46$ (Figure F.1a). For most of sampling, including the steps that fix the layout, Constancy is thus the NTK base with a log-N temperature: the gate is principled, but at 4096^2 the bands it guards stay below the noise floor until late.

A native-resolution target trades sharpness for spectral match. The time-matched target opens the gate early. On the held-out Constancy-TM runs, corrections exceed 0.1 in log gain after the first 8 steps (median), and 70% of all nonzero corrections are negative: they lower the mid pairs 6 to 8 (−0.24 at the last step) and, late, the finest pairs 0 to 2 (−0.22). Matching the native spectrum at each noise level thus damps the bands that carry 4K detail, and the held-out outcome follows: against SEGA base + temp., the spectral errors shrink slightly while both MUSIQ scores fall. Spectral agreement with native statistics and sharpness as MUSIQ measures it are different objectives at this resolution.

SEGA's allocation is mostly a fixed tilt. Across the held-out prompts the released logit gain stays between 1.00 and 1.61. A fixed tilt over pairs explains 56% of its variance and a tilt plus a per-step profile 80%. The prompt-dependent remainder, 20% overall and 7.3% from step 14 on, is an upper bound on content dependence, because it also contains each prompt's noise draw. At the last step the finest pair has a logit gain of 1.47 and the 6 slowest pairs per axis sit at the clamp of 1. A law that adapts more to content therefore has little to copy: SEGA + TM raises the prompt-dependent share to 37% and gains nothing held out. A CPU study of the printed equations finds the same structure on another latent space (Appendix F).

The actuator is weak and prompt-dependent. Every law assumes that raising a band's gain raises its output power. A probe on three dev prompts that scales one octave's gain by 1.5 (Figure F.1b) finds a clear same-octave width-axis response for one prompt and a near-zero one for the other two, off-diagonal responses spread across octaves, and a weak height-axis response. On two dev prompts, a local linear model predicts half-step band-power changes little better than no change (Appendix F). Gain is a blunt, prompt-dependent handle on band power; a per-band law would need a measured response model of the network, not only a target for the image.

5.3 Results depend on the checkpoint

Our first campaign used FLUX.1-dev, on which DyPE and Constancy produced a dense halftone lattice over the 4096^2 frame and NTK a featureless blur. To tell a bug in our port from a property of the model, we ran the official DyPE code unchanged. It produces the same lattice on FLUX.1-dev and a clean image on FLUX.1-Krea-dev (Figure 4), the checkpoint that the official DyPE and SEGA scripts load. We moved every experiment to FLUX.1-Krea-dev and discarded the FLUX.1-dev pilot.

Mechanism measurements need not transfer either: on one dev prompt, raising the gain of the fastest octave sharply lowers that band's power on FLUX.1-dev and slightly raises it on FLUX.1-Krea-dev, although guidance and code revision also differ between the two probes (Appendix F).

5.4 SEGA: paper versus code

With the printed law (Equation (2) and Table 2), the allocation is unbounded, and at 4096^2 the physical bin map sends every pair with a wavelength above 256 tokens, 19 of the 28 per axis, to the same

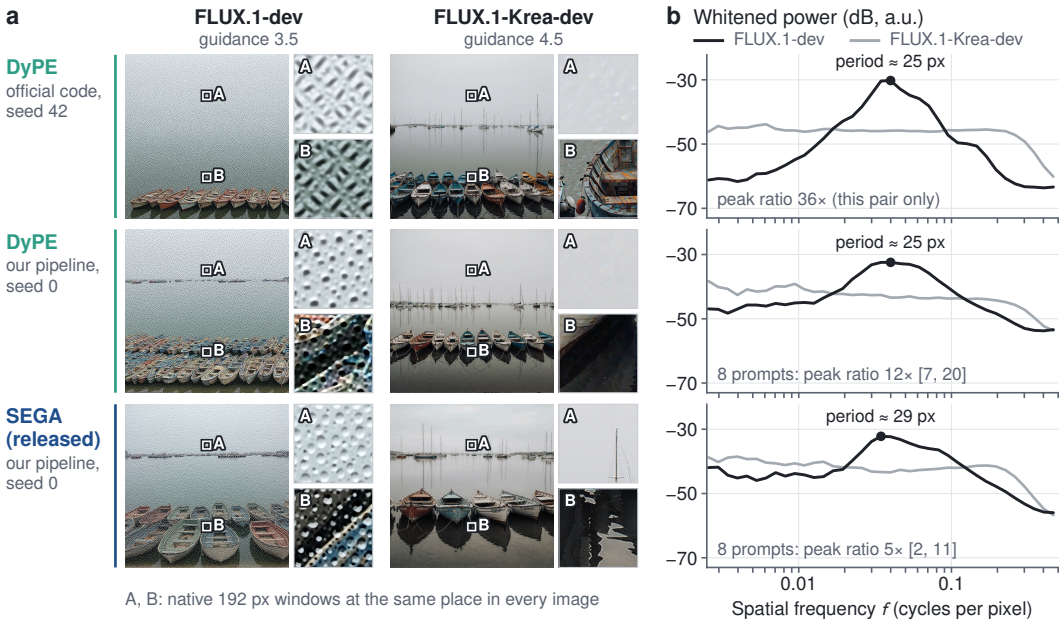


Figure 4: FLUX.1-dev breaks into a dot lattice at 4096^2 even under the official DyPE code, while FLUX.1-Krea-dev shows none; results on one checkpoint do not transfer. (a) Dev prompt 0 under the official DyPE code (seed 42) and our DyPE and SEGA (released) pipelines (seed 0), at guidance 3.5 (FLUX.1-dev) and 4.5 (FLUX.1-Krea-dev); A and B are native 192-pixel windows at the same place in every image. (b) Whitened radial power spectra of luminance for each row: FLUX.1-dev peaks at the lattice period and FLUX.1-Krea-dev does not; peak ratios are given for this pair (top) and over the eight dev prompts (descriptive).

frequency bin. The released law scales the spread by 0.15 and clamps the result at 1, and its logit gain stays between 1.00 and 1.61 on our held-out runs. In the eight-prompt dev pilot (descriptive only; Appendix D.3), our implementation of the printed equations scores worse than the released code on every metric; freezing the printed allocation into a static tilt destroys the images, while freezing the released allocation costs much less (Figure A.2). Readers who reimplement SEGA from the paper will not reproduce the released method, and comparisons against SEGA should state which version they ran.

5.5 Pitfalls in 4K evaluation

Besides the flow shift of Section 2, two details changed what our numbers can show or what they cost.

224-pixel scorers miss the finest octaves. CLIP, PickScore and ImageReward see a 224-pixel view, whose Nyquist frequency is 0.0273 cycles per original pixel ($2^{-5.2}$); bands 7 to 10 of our 11 octave bands lie entirely above it. These are the bands where the arms differ (Figure A.3 and Table B.4): in band 7, NTK falls far below the reference profile (log-power error −1.30), the two arms with SEGA's allocation stay closest to it (SEGA (released) +0.06, SEGA + TM −0.01), and SEGA base + temp. carries excess power (+0.39). This is why SEGA base + temp. can tie SEGA (released) on all three reward models while trailing it on full-frame MUSIQ, and why a 4K comparison that reports only these scorers cannot detect a change in fine detail.

Attention kernel. On the same GH200 node and method, forcing FlashAttention-2 makes a 4096^2 sample take 1.54 times as long as cuDNN-first SDPA (medians 290 s and 188 s), while method families on one node differ by at most 0.6%; one GB200 GPU is 2.01 times faster than a GH200 (Table G.1). Timings of different methods are comparable only with the kernel and the GPU fixed. Our records did not log the dispatched kernel or a hash of the loaded weights; we recovered both from code revisions and task scripts (Appendix G).

6 Discussion and limitations

What to report. A method that changes the rotary base, the temperature and per-band gains at once should report a control that keeps its base and temperature and drops the rest; here that control leaves a gap of 2.09 full-frame MUSIQ points and none on the reward models. Claims that credit spectral or content adaptivity should be tested against this control with full-resolution metrics, because 224-pixel scorers cannot see where such methods act.

Small dev sets mislead. Eight dev prompts were enough to reject clearly worse variants and too few for anything else. On the dev set SEGA base + temp. looked slightly better than SEGA (released) on MUSIQ, yet held out it trails by 2.09 points; the one variant that passed the second gate lost on PickScore and crop MUSIQ held out; and a rerun of SEGA (released) on the same dev prompts moved MUSIQ by more than the margin of the one variant that passed the third gate (Appendix D.2). The held-out set has now been inspected and cannot serve as fresh confirmation for new variants.

Limitations. We study one model family (FLUX.1-Krea-dev) at one square size, 4096^2; other aspect ratios, sizes, backbones and video are untested, and Section 5.3 suggests that results may not transfer. All metrics are automatic: the reward models see 224 pixels, MUSIQ and CLIP-IQA are learned proxies with their own biases, the spectral error uses a mean reference profile rather than a per-prompt target, and we ran no human study. Most arms have one seed; the seed-1 replication covers two arms on 55 prompts. Intervals are per metric and not adjusted for multiple comparisons; Table B.3 alone holds 60. The original Constancy has only a partial held-out run. The paper-versus-code numbers come from an eight-prompt dev pilot and are descriptive, the mechanism probes use three dev prompts or fewer, and the CPU study of SEGA uses another latent space. We port SEGA and DyPE from their released code with parity tests and ran the official DyPE code only for Section 5.3. A 4096^2 sample takes a median of 188 s on a GH200 (Appendix G), which limited the number of arms, seeds and resolutions.

7 Related work

Training-free high-resolution generation. U-Net methods dilate convolutions in space [22] or frequency [26], fuse patches [14] or scales [42], restructure features [56] or cascade over frequencies [57]. For rectified-flow transformers [5, 16, 40], I-Max [15] and HiFlow [8] guide the high-resolution trajectory with a low-resolution one, HRDiT [54] adapts pretrained transformers, and Diffusion-4K [55] fine-tunes at 4K and supplies our prompts. We study the single-pass, embedding-level family, which changes no weights and adds no passes.

Rotary extrapolation and attention temperature. Language models remap rotary frequencies by interpolation [10], NTK-aware scaling [7], YaRN [41] or LongRoPE [12]; Barbero et al. [3] and DoPE [52] analyse which frequencies carry position and which carry semantics. For diffusion transformers, RIFLEx [58] lowers the frequency that causes repetition in long videos, DyPE [27] schedules the remapping with the noise level, UltraImage [59] revisits frequency selection, TIDE [36] adds step-aware temperature control, SEGA [45] adds per-band gains, and AReA [44], from SEGA's authors, adapts attention locality. Scaling logits by the log of the key count keeps attention entropy roughly constant, as in variable-size diffusion [28] and Scalable-Softmax [37]; our results suggest that this knob, together with the base, does most of the work at 4096^2.

Spectral views and efficient coding. Diffusion denoises roughly coarse to fine [11, 17], and denoisers adapt to image structure through harmonic bases [21, 29]. Spectral guidance steers toward target spectra [31], CSFlow weights flow matching by contrast sensitivity [19], and Attention Frequency Modulation edits the Fourier bands of cross-attention [38]. Our controllers act on the rotary bands of self-attention and draw on natural-image spectra [13, 18], human sensitivity [33, 39], efficient coding [1, 2, 48], contrast constancy [20] and blur adaptation by divisive gain control [9, 23, 46, 51].

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A Additional figures

Figure A.1 shows held-out samples, Figure A.2 the dev pilot of SEGA's paper and code, and Figure A.3 the octave-band spectra discussed in Section 5.5.

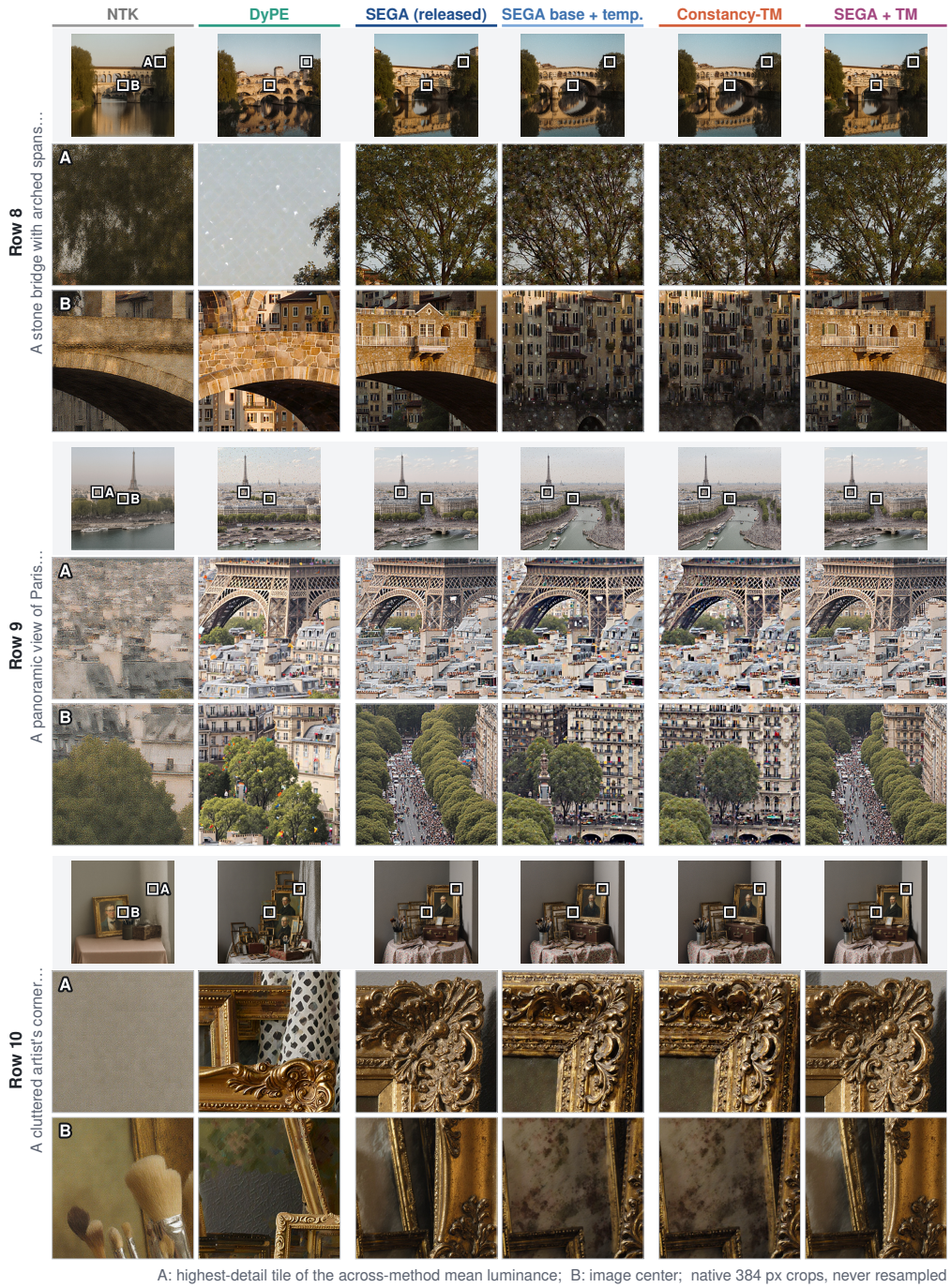


Figure A.1: At 4096^2 NTK loses fine detail, while the other arms render comparable detail. Held-out rows 8 to 10, the first three held-out rows, fixed before any output was viewed; seed 0, one sample per arm. Full frames are Lanczos-reduced; native-resolution crops sit at identical coordinates (A: highest-detail tile of the across-arm mean luminance; B: image centre) and are never resampled. Compositions differ between arms, so a crop can show different content. DyPE shows faint speckle in flat sky on held-out row 8 (one crop checked).

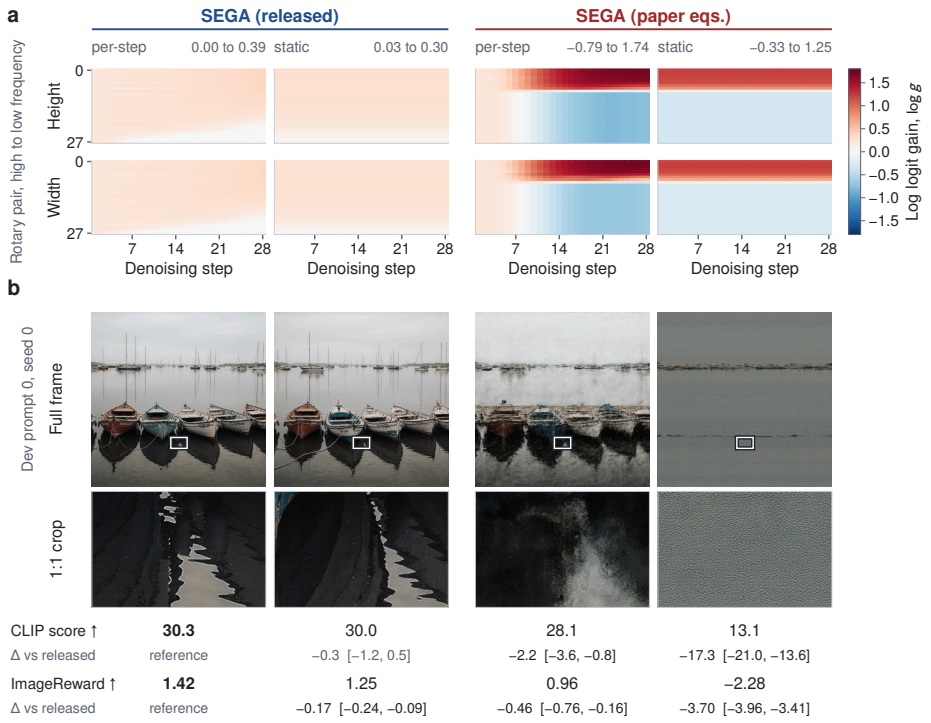


Figure A.2: SEGA as printed is not SEGA as released: the printed equations swing the logit gain much more widely and degrade the images, and their static tilt destroys them. Eight-prompt dev pilot (rows 0–7, seed 0), descriptive only. (a) Median log logit gain per rotary pair and denoising step over the eight dev prompts, on one shared scale (ranges printed above the panels); static variants average each method's gains over prompts and steps. (b) Dev prompt 0, seed 0: full frames and a native-resolution crop, above CLIP score and ImageReward means over the eight dev prompts with paired differences from SEGA (released) and 95% bootstrap intervals.

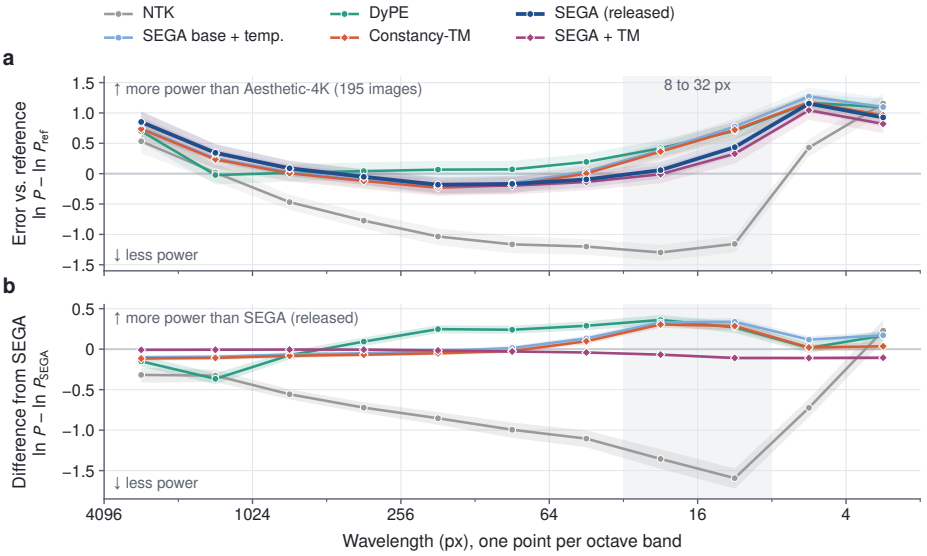


Figure A.3: Without SEGA's per-band allocation, luminance power at 8 to 32 pixel wavelengths sits above released SEGA's; Constancy-TM leaves a similar excess, and NTK falls short from 8 to 1024 pixels. (a) Signed log-power error against the mean ln P of the 195 Aesthetic-4K references, per octave band of linear-sRGB luminance; (b) paired difference from SEGA (released). Held-out rows 8–194 (n = 187, seed 0); bands are 95% bootstrap intervals over prompts. The shaded 8 to 32 pixel band lies above the Nyquist frequency of a 224-pixel view.

B Full held-out results

Table B.1 decomposes SEGA's gain over NTK with means and paired gains, Table B.2 gives absolute means and Table B.3 the paired differences for all ten metrics. The spectral rows use the mean octave profile of the 195 Aesthetic-4K reference images as the target; Table B.4 breaks the spectral error into its 11 octave bands. FID and KID are not tabulated: for the seed-1 images (56 prompts per arm, including one dev row), the paired FID difference of SEGA base + temp. against SEGA (released) is +3.26 with interval [−6.86, +13.87], and that of Constancy-TM is +3.74 with interval [−6.72, +14.08].

Table B.1: Where SEGA's improvement over NTK comes from. Means on the same 187 held-out prompts, paired gains from base and temperature and from the allocation, and the share of the total gain carried by base and temperature, (SEGA base + temp. − NTK)/(SEGA (released) − NTK), with its 95% paired bootstrap interval. A share above 100% means the allocation's point estimate is negative.

Metric	Mean (n = 187)			Paired gain		
	NTK	SEGA base + temp.	SEGA (released)	base + temp.	alloc-ation	share of base + temp.
ImageReward↑	0.596	1.216	1.190	+0.620 ▲ [+0.525, +0.724]	−0.026 [−0.070, +0.016]	104% [97, 113]
PickScore↑	22.03	23.08	23.05	+1.05 ▲ [+0.92, +1.18]	−0.03 [−0.09, +0.03]	103% [97, 109]
CLIP↑	27.15	29.15	29.14	+2.01 ▲ [+1.57, +2.48]	−0.02 [−0.23, +0.19]	101% [90, 112]
MUSIQ↑	29.49	43.14	45.23	+13.65 ▲ [+12.05, +15.28]	+2.09 ▲ [+0.96, +3.25]	87% [80, 94]
MUSIQcrop↑	39.63	58.22	58.33	+18.59 ▲ [+16.72, +20.63]	+0.11 [−0.51, +0.71]	99% [96, 103]
CLIP-IQA↑	0.557	0.594	0.603	+0.037 ▲ [+0.021, +0.053]	+0.009 ▲ [+0.001, +0.018]	80% [59, 98]

Table B.2: Held-out means with 95% bootstrap intervals (rows 8–194, seed 0). Constancy is the partial run on 41 prompts (Appendix D).

Metric	SEGA (released) n = 187	NTK n = 187	SEGA base + temp. n = 187	DyPE n = 187	Constancy n = 41	Constancy-TM n = 187	SEGA + TM n = 187
ImageReward↑	1.190 [1.087, 1.284]	0.596 [0.456, 0.736]	1.216 [1.118, 1.313]	1.026 [0.910, 1.142]	1.177 [0.953, 1.392]	1.208 [1.110, 1.304]	1.199 [1.096, 1.293]
PickScore↑	23.05 [22.89, 23.21]	22.03 [21.84, 22.21]	23.08 [22.92, 23.24]	22.57 [22.36, 22.75]	22.70 [22.29, 23.08]	23.01 [22.84, 23.16]	22.98 [22.81, 23.13]
CLIP↑	29.14 [28.70, 29.55]	27.15 [26.64, 27.64]	29.15 [28.72, 29.58]	28.76 [28.32, 29.20]	29.46 [28.40, 30.49]	28.87 [28.42, 29.31]	29.04 [28.61, 29.47]
MUSIQ↑	45.23 [43.69, 46.79]	29.49 [28.13, 30.95]	43.14 [41.58, 44.61]	43.35 [41.71, 45.02]	40.55 [37.20, 43.56]	42.19 [40.65, 43.58]	45.02 [43.56, 46.53]
MUSIQcrop↑	58.33 [57.06, 59.63]	39.63 [37.77, 41.33]	58.22 [57.07, 59.44]	60.30 [59.12, 61.53]	60.75 [58.19, 63.01]	56.14 [54.98, 57.36]	57.20 [55.84, 58.52]
CLIP-IQA↑	0.603 [0.590, 0.616]	0.557 [0.546, 0.568]	0.594 [0.581, 0.607]	0.632 [0.618, 0.645]	0.572 [0.544, 0.601]	0.578 [0.566, 0.590]	0.606 [0.593, 0.619]
CLIP-IQAcrop↑	0.586 [0.573, 0.600]	0.537 [0.525, 0.549]	0.569 [0.556, 0.583]	0.599 [0.585, 0.613]	0.543 [0.514, 0.572]	0.550 [0.538, 0.563]	0.588 [0.576, 0.602]
Spec. err.↓	1.068 [1.005, 1.133]	1.285 [1.214, 1.358]	1.102 [1.035, 1.168]	1.096 [1.024, 1.168]	1.337 [1.205, 1.477]	1.088 [1.020, 1.152]	1.066 [1.001, 1.133]
Slope err.↓	0.200 [0.180, 0.221]	0.165 [0.146, 0.185]	0.224 [0.200, 0.247]	0.218 [0.194, 0.242]	0.276 [0.218, 0.338]	0.218 [0.195, 0.240]	0.197 [0.178, 0.218]
Tiling↓	0.0684 [0.0568, 0.0802]	0.0757 [0.0612, 0.0905]	0.0616 [0.0507, 0.0732]	0.0865 [0.0778, 0.0953]	0.0589 [0.0390, 0.0797]	0.0616 [0.0505, 0.0732]	0.0684 [0.0566, 0.0804]

C Seed-1 replication

Table C.1 repeats the comparison for SEGA base + temp. and Constancy-TM on held-out rows 8–62 with a second seed, next to seed 0 on the same 55 prompts. One of the verdicts that changes, crop CLIP-IQA for SEGA base + temp. at seed 1, rests on an interval that excludes zero by less than 10^{-4}, so the agreement count can move by one under another bootstrap seed. Every verdict that changes moves between a tie and a one-sided verdict. Spectral log-RMSE differences grow at seed 1 (SEGA base + temp.: +0.080 against +0.056 at seed 0 on the same prompts; Constancy-TM: +0.062 against +0.042).

Table B.3: Held-out paired differences against SEGA (released), all metrics. Format as in Table 3. Constancy is the partial run on 41 prompts (Appendix D).

Metric	NTK $n = 187$	SEGA base + temp. $n = 187$	DyPE $n = 187$	Constancy $n = 41$	Constancy-TM $n = 187$	SEGA + TM $n = 187$
ImageReward↑	−0.594 ▾ [−0.701, −0.494]	+0.026 [−0.016, +0.070]	−0.164 ▾ [−0.258, −0.087]	+0.018 [−0.089, +0.133]	+0.018 [−0.023, +0.059]	+0.009 [−0.008, +0.027]
PickScore↑	−1.02 ▾ [−1.16, −0.89]	+0.03 [−0.03, +0.09]	−0.49 ▾ [−0.63, −0.36]	−0.07 [−0.28, +0.18]	−0.05 [−0.11, +0.02]	−0.08 ▾ [−0.10, −0.05]
CLIP↑	−1.99 ▾ [−2.43, −1.55]	+0.02 [−0.19, +0.23]	−0.37 ▾ [−0.72, −0.01]	−0.05 [−0.78, +0.68]	−0.27 ▾ [−0.49, −0.04]	−0.09 ▾ [−0.20, −0.00]
MUSIQ↑	−15.74 ▾ [−17.50, −14.00]	−2.09 ▾ [−3.25, −0.96]	−1.88 ▾ [−3.26, −0.56]	−5.09 ▾ [−8.44, −1.74]	−3.04 ▾ [−4.28, −1.81]	−0.21 [−0.71, +0.32]
MUSIQcrop↑	−18.70 ▾ [−20.68, −16.86]	−0.11 [−0.71, +0.51]	+1.97 ▴ [+1.16, +2.80]	+1.91 ▴ [+0.10, +3.80]	−2.19 ▾ [−2.81, −1.57]	−1.13 ▾ [−1.39, −0.90]
CLIP-IQA↑	−0.046 ▾ [−0.062, −0.031]	−0.009 ▾ [−0.018, −0.001]	+0.029 ▴ [+0.020, +0.038]	−0.005 [−0.031, +0.019]	−0.025 ▾ [−0.034, −0.017]	+0.003 [−0.000, +0.006]
CLIP-IQAcrop↑	−0.049 ▾ [−0.066, −0.034]	−0.017 ▾ [−0.026, −0.008]	+0.013 ▴ [+0.003, +0.023]	−0.027 [−0.053, +0.001]	−0.036 ▾ [−0.045, −0.027]	+0.002 [−0.001, +0.006]
Spec. err.↓	+0.216 ▾ [+0.154, +0.287]	+0.034 ▾ [+0.016, +0.051]	+0.028 [−0.007, +0.061]	+0.204 ▾ [+0.127, +0.277]	+0.019 ▾ [+0.001, +0.037]	−0.002 [−0.009, +0.004]
Slope err.↓	−0.035 ▴ [−0.053, −0.018]	+0.024 ▾ [+0.013, +0.034]	+0.018 ▾ [+0.003, +0.033]	+0.061 ▾ [+0.021, +0.101]	+0.018 ▾ [+0.008, +0.027]	−0.003 [−0.007, +0.002]
Tiling↓	+0.0074 [−0.0044, +0.0185]	−0.0068 ▴ [−0.0108, −0.0026]	+0.0182 ▾ [+0.0085, +0.0279]	−0.0099 [−0.0223, +0.0041]	−0.0068 ▴ [−0.0107, −0.0026]	+0.0001 [−0.0007, +0.0008]

Table B.4: Spectral error by octave band. Mean log-power difference from the Aesthetic-4K reference profile on held-out prompts; positive values mean excess power. Bands 7 to 10 lie above the Nyquist frequency of a 224-pixel view of a 4096^2 image. Constancy is the partial run on 41 prompts (Appendix D).

Octave band	0	1	2	3	4	5	6	7	8	9	10
Centre (cyc/px)	$2^{-11.5}$	$2^{-10.5}$	$2^{-9.5}$	$2^{-8.5}$	$2^{-7.5}$	$2^{-6.5}$	$2^{-5.5}$	$2^{-4.5}$	$2^{-3.5}$	$2^{-2.5}$	$2^{-1.5}$
SEGA (released)	+0.85	+0.34	+0.09	−0.05	−0.18	−0.17	−0.10	+0.06	+0.44	+1.15	+0.93
NTK	+0.53	+0.02	−0.47	−0.77	−1.03	−1.16	−1.20	−1.30	−1.16	+0.43	+1.15
SEGA base + temp.	+0.75	+0.25	+0.02	−0.10	−0.21	−0.15	+0.04	+0.39	+0.78	+1.27	+1.10
DyPE	+0.70	−0.02	+0.01	+0.04	+0.07	+0.07	+0.19	+0.42	+0.71	+1.17	+1.09
Constancy	+0.73	+0.34	+0.27	+0.23	+0.21	+0.36	+0.72	+1.37	+1.72	+1.84	+1.64
Constancy-TM	+0.74	+0.24	+0.01	−0.12	−0.23	−0.19	+0.00	+0.36	+0.72	+1.17	+0.96
SEGA + TM	+0.84	+0.34	+0.08	−0.06	−0.20	−0.20	−0.14	−0.01	+0.33	+1.04	+0.82

Table C.1: Seed 1 repeats the seed-0 verdicts. Paired differences against SEGA (released) on the same 55 held-out prompts (rows 8–62) at seed 0 and at seed 1. 17 of 20 verdicts agree. Format as in Table 3.

	SEGA base + temp. − SEGA (released)		Constancy-TM − SEGA (released)	
Metric	seed 0	seed 1	seed 0	seed 1
ImageReward↑	+0.070 [−0.011, +0.160]	+0.031 [−0.036, +0.094]	+0.063 [−0.014, +0.142]	+0.019 [−0.045, +0.083]
PickScore↑	+0.05 [−0.07, +0.20]	+0.00 [−0.12, +0.13]	−0.01 [−0.12, +0.12]	−0.08 [−0.20, +0.05]
CLIP↑	−0.02 [−0.54, +0.47]	+0.02 [−0.40, +0.46]	−0.31 [−0.82, +0.17]	−0.32 [−0.78, +0.13]
MUSIQ↑	−3.62 ▾ [−5.84, −1.52]	−2.79 ▾ [−4.58, −0.95]	−4.36 ▾ [−6.57, −2.32]	−3.53 ▾ [−5.38, −1.69]
MUSIQcrop↑	+0.43 [−0.53, +1.40]	+0.35 [−0.69, +1.44]	−1.77 ▾ [−2.82, −0.77]	−1.99 ▾ [−3.09, −0.91]
CLIP-IQA↑	+0.002 [−0.012, +0.017]	−0.008 [−0.023, +0.008]	−0.018 ▾ [−0.034, −0.002]	−0.025 ▾ [−0.043, −0.007]
CLIP-IQAcrop↑	−0.008 [−0.024, +0.009]	−0.015 ▾ [−0.032, −0.000]	−0.034 ▾ [−0.050, −0.015]	−0.042 ▾ [−0.059, −0.025]
Spec. err.↓	+0.056 ▾ [+0.020, +0.091]	+0.080 ▾ [+0.042, +0.120]	+0.042 ▾ [+0.006, +0.077]	+0.062 ▾ [+0.026, +0.101]
Slope err.↓	+0.035 ▾ [+0.018, +0.053]	+0.028 ▾ [+0.005, +0.050]	+0.029 ▾ [+0.013, +0.046]	+0.025 ▾ [+0.004, +0.046]
Tiling↓	−0.0097 ▴ [−0.0157, −0.0038]	−0.0055 [−0.0136, +0.0027]	−0.0098 ▴ [−0.0156, −0.0040]	−0.0058 [−0.0139, +0.0023]

D Partial and dev-only runs (not held-out results)

Nothing in this appendix is a held-out result of the main protocol. The first subsection reports a held-out run that covers only part of the held-out set and started before the protocol was fixed; the other two report dev-prompt selection evidence and a descriptive pilot.

D.1 Partial held-out run of Constancy

The original Constancy was generated on 41 of the 187 held-out prompts (seed 0) before the protocol was fixed (Appendix D.2). Every number in this subsection comes from those 41 prompts only, and differences are paired on them. Against SEGA (released) the run ties on the reward models, loses 5.09 full-frame MUSIQ points, gains crop MUSIQ (+1.91) and has a much larger spectral error

(+0.204), with excess power in the top octaves and the largest excess of any arm in band 8 (log-power error +1.72; Table B.4). Measured as a share of the gain of SEGA (released) over NTK on the same prompts ($n = 41$), it recovers 110% [100, 120] on crop MUSIQ but 68% [47, 88] on full-frame MUSIQ.

The visibility gate of Section 5.2 opens late on these prompts as on the dev prompts. On the 41 held-out prompts it opens only after the first 20 of 28 steps (median; range 16 to 22), at $\sigma = 0.56$ (range 0.70 to 0.46), and the correction then acts in a median of 8 steps.

The run also separates the temperature from the base. NTK and SEGA base + temp. differ in both knobs, but Constancy adds to the NTK base only the log-N temperature ($\tau = 1.333$) and a per-band correction that, as the onset statistics above show, stays at zero for the first 20 of 28 steps (median). On its 41 prompts it improves on NTK by +0.688 [+0.479, +0.917] ImageReward and +21.66 [+17.43, +26.06] crop MUSIQ; on the same prompts, the entire SEGA (released) minus NTK gap is +0.670 ImageReward and +19.75 crop MUSIQ. On this partial run, most of the base-and-temperature share is therefore temperature.

D.2 Dev selection

Table D.1 lists every variant scored on the eight dev prompts, with its mean and its difference from the SEGA (released) reference of the same campaign. These numbers chose what to run held-out; they are not results. Campaign 1 preceded the written gate: it sent SEGA base + temp. to the held-out set as a decomposition control and Constancy-TM as the candidate, and seven chunks of the original Constancy had already run on held-out rows when the protocol was fixed, which is why that arm has only 41 held-out prompts. Campaigns 2 and 3 applied the gate of Section 4. In campaign 3 a fresh SEGA (released) rerun served as the reference. It differs from the pilot run of the same prompts and seed by +0.79 MUSIQ and −0.032 ImageReward, which is more than the MUSIQ margin by which the one passing variant cleared the gate (+0.39); we did not establish the source of the difference. That variant was not run on held-out prompts, and we make no claim about it. The dev set also misjudged the allocation itself: on the eight dev prompts SEGA base + temp. scored +0.24 MUSIQ against SEGA (released), while held out it trails by 2.09.

D.3 SEGA paper versus code: pilot numbers

In an eight-prompt pilot on the dev prompts, which is descriptive only, our implementation of the printed equations scores ImageReward 0.964, CLIP 28.14 and full-frame MUSIQ 30.78, against 1.423, 30.32 and 42.49 for the released code (Table D.2). Freezing the printed allocation into a static tilt, a natural ablation of content adaptivity, destroys the images: CLIP falls to 13.06 and logit gains reach 3.49. Freezing the released allocation instead costs much less (CLIP 29.98, ImageReward 1.255). Figure A.2 in Appendix A shows the two gain trajectories and the resulting images. The pilot also ran YaRN, which follows a literal reading of a design note rather than the usual rotation-count convention.

E Implementation details

Pipeline. The FLUX.1-Krea-dev transformer, loaded through an environment override (`CROPE_TRANSFORMER`), runs inside the FLUX.1-dev pipeline (T5 and CLIP text encoders with 512 T5 tokens, autoencoder with tiled decoding at 4096^2), diffusers 0.39 [49], PyTorch 2.11, bfloat16, batch size one, a per-prompt CPU seed, 28 Euler steps and distilled guidance 4.5. The transformer has 19 double-stream and 38 single-stream blocks with 24 heads of dimension 128, and normalizes q and k before the rotation. Every arm multiplies q and k of every token in every block by one gain vector after the rotation; the vector is shared by blocks and heads and recomputed once per model call. The scheduler patch that holds $\mu = 1.15$ is applied before any configuration is recorded. The per-image JSON records hard-code the base model name and an attention label; the loaded transformer comes from the task scripts and the attention policy from the code revision (Section 5.5 and Appendix G).

DyPE. For spatial axis a with scale s and $\kappa = \sigma^2$, let $b(r) = D \ln\big(64/(2\pi r)\big)/(2 \ln \theta)$ be the pair index that completes r rotations over 64 tokens and $\mu_{r_1,r_2}(j) = 1 - \mathrm{clip}\big((j-\ell)/(h-\ell), 0, 1\big)$ with

Table D.1: Dev-set selection (rows 0–7, seed 0, $n = 8$); selection evidence only. Each cell: mean, and below it the difference from the campaign's SEGA (released) reference. [a]Partial held-out run (41 prompts) started before the dev decision. [b]Fresh rerun; the second line is its difference from the pilot run. [c]Passed the dev gate; never evaluated on held-out prompts.

Variant	Image-Reward↑	MUSIQ full↑	CLIP score↑	Pick-Score↑	MUSIQ crop↑	Outcome
Campaign 1 (reference: SEGA (released), pilot run)						
SEGA (released)	1.423 +0.000	42.49 +0.00	30.32 +0.00	23.41 +0.00	58.23 +0.00	reference
NTK + log-N temp.	1.282 −0.141	40.36 −2.13	29.96 −0.36	23.12 −0.29	60.29 +2.05	dev only
Constancy	1.285 −0.138	40.56 −1.93	30.10 −0.22	23.10 −0.31	58.62 +0.39	partial held-out[a]
Constancy-TM on NTK base, $\eta = 1$	1.314 −0.109	41.87 −0.62	30.04 −0.28	23.10 −0.31	57.51 −0.72	dev only
Constancy-TM on NTK base, $\eta = 2$	1.313 −0.110	43.34 +0.85	30.13 −0.19	23.04 −0.37	54.61 −3.63	dev only
SEGA base + temp.	1.444 +0.021	42.72 +0.24	30.32 +0.00	23.26 −0.15	55.93 −2.30	held-out (control)
Constancy-TM	1.432 +0.009	41.65 −0.84	30.24 −0.08	23.25 −0.16	54.57 −3.66	held-out (candidate)
Campaign 2, written gate (reference: SEGA (released), pilot run)						
Alias-T	1.281 −0.142	35.81 −6.68	29.37 −0.95	23.06 −0.35	63.66 +5.43	fails gate
Alias-T-s	1.277 −0.146	44.97 +2.48	30.24 −0.08	23.47 +0.06	58.73 +0.49	fails gate
SEGA + TM, $\eta = 0.5$	1.441 +0.018	42.48 −0.01	30.20 −0.12	23.31 −0.10	58.51 +0.27	fails gate
SEGA + TM, $\eta = 1$	1.449 +0.026	42.51 +0.02	30.34 +0.02	23.33 −0.08	58.28 +0.04	passes, held-out
Campaign 3, same gate (reference: SEGA (released), fresh rerun)						
SEGA (released) (rerun)	1.391 −0.032	43.28 +0.79	30.15 −0.17	23.38 −0.03	58.59 +0.36	reference[b]
SEGA-RMS	1.428 +0.036	43.67 +0.39	30.08 −0.07	23.31 −0.07	58.38 −0.21	passes, not confirmed[c]
SEGA-Uniform	1.374 −0.017	41.71 −1.57	30.73 +0.59	23.29 −0.08	58.07 −0.52	fails gate
SEGA + TM-lag	1.411 +0.019	43.06 −0.22	30.16 +0.01	23.32 −0.05	58.44 −0.15	fails gate

Table D.2: Eight-prompt pilot on the dev prompts (rows 0–7, seed 0), descriptive only. Means; colour marks a paired bootstrap interval against SEGA (released) that excludes zero, which with $n = 8$ is indicative at best. Static variants freeze the allocation instead of reading the current latent.

Method	Image-Reward↑	Pick-Score↑	CLIP score↑	MUSIQ full↑	MUSIQ crop↑	CLIP-IQA full↑	Spectral error↓
SEGA (released)	1.423	23.41	30.32	42.49	58.23	0.557	0.857
SEGA (released, static)	1.255 ▾	23.12	29.98	43.25	57.75	0.559	0.870
SEGA (paper eqs.)	0.964 ▾	22.18 ▾	28.14 ▾	30.78 ▾	42.73 ▾	0.455 ▾	0.944
SEGA (paper eqs., static)	−2.277 ▾	16.93 ▾	13.06 ▾	28.49 ▾	35.53 ▾	0.435 ▾	4.369 ▾
NTK	0.757 ▾	22.37 ▾	27.84 ▾	32.91 ▾	41.41 ▾	0.543	1.220 ▾
YaRN	−0.163 ▾	21.14 ▾	28.06 ▾	42.51	66.69 ▴	0.645 ▴	1.250 ▾
DyPE	1.118 ▾	22.23 ▾	28.59 ▾	44.38	58.32	0.584	0.920
NTK + log-N temp.	1.282 ▾	23.12	29.96	40.36	60.29	0.578	1.038 ▾
Constancy	1.285 ▾	23.10	30.10	40.56	58.62	0.563	1.017 ▾

$\ell = \max(\lfloor b(r_1) \rfloor, 0)$ and $h = \min(\lceil b(r_2) \rceil, D/2)$. Then

$$\omega_j^{\text{mix}} = \tfrac{\omega_j}{s}\big(1 - \mu_{1.25\kappa,0.75\kappa}(j)\big) + \omega_j^{\text{ntk}}\,\mu_{1.25\kappa,0.75\kappa}(j), \qquad \omega'_j = \omega_j^{\text{mix}}\big(1 - \mu_{16\kappa,2\kappa}(j)\big) + \omega_j\,\mu_{16\kappa,2\kappa}(j),$$

with ω^{ntk} the NTK-aware frequencies, and the spatial cos/sin amplitude is $1 + 0.1 \ln s$, as in the official code at the inspected revision.

SEGA (released). The sensor takes the channel mean Z of the projected input features on the token grid. The axis profile E_a is the one-dimensional FFT power along axis a averaged over the other axis and rebinned into $K_a = \max(\lfloor L_a/2 \rfloor, 8)$ bins. Pair j reads $\ell_j = \ln E_a\big((1 - \rho_j)(K_a - 1)\big)$ by linear interpolation, with $\rho_j = j/(D/2 - 1)$ the normalized rank of its log period; z_j standardizes ℓ_j with the sample standard deviation; and $m_{a,j} = \max\big(1, m_{\text{ref}}[1 - 0.15\,\varsigma\,c_j]\big)$ with $c_j = \tanh(1.5 z_j) - \text{mean}_i \tanh(1.5 z_i)$, $m_{\text{ref}} = \max(s, 1)^{0.08}$ and $\varsigma = 1 - \text{SF}^{1.5}$ from the radially binned two-dimensional power. The index axis keeps gain 1.

Constancy. Band energies are densities: with an orthonormal FFT, $P(k) = \sum_c |\mathcal{F}(x_c - \bar{x}_c)(k)|^2$, $E_a(k)$ averages P over the other axis, and $E_a[b]$ averages over half-octave bands b (13 bands at 256 tokens per side). Unit white noise gives $E_a[b] = C = 64$ in expectation, so the noise floor at level

σ is $\sigma^2 C$. The gate uses the current input; the error uses the clean estimate of the previous call, $\hat{x} = x_{n-1} - \sigma_{n-1} v_\theta(x_{n-1})$. Corrections are interpolated in $\log f$ between band centres and applied to pairs with frequency above one cycle per grid, which at 4096^2 are pairs 0 to 8 of each axis. Step 0 has no clean estimate and uses the temperature alone, and $|\ln g - \ln \tau| \le \eta$ always. The native targets come from 1024^2 samples of the same model on 16 Pick-a-Pic prompts [32] disjoint from the evaluation set: the late target averages the three steps with $\sigma < 0.3$, and the time-matched target keeps all 28 steps and is interpolated in σ.

Alias-T. Alias-T applies the log-N temperature of Section 2 pair by pair. A pair of wavelength $\lambda_{a,j}$ tokens under the released base, on a grid of L tokens per side ($L = 64$ at native size), gets the logit gain

$$g_{a,j} = \frac{\ln\big(n(\lambda_{a,j}, L)\, L\big)}{\ln\big(n(\lambda_{a,j}, 64) \cdot 64\big)}, \qquad n(\lambda, L) = \min\big(L, \max(1, L/\lambda)\big),$$

where $n(\lambda, L)$, the pair's alias count, is the number of its wavelengths across the grid, bounded by 1 and L; the index axis takes τ_N. The gains are static and content-free, with no correction and no further temperature. Alias-T-s shifts every $\ln g$ by one constant so that the mean over all 128 dimensions equals SEGA's reference log gain, $0.16 \ln s$.

F Mechanism diagnostics

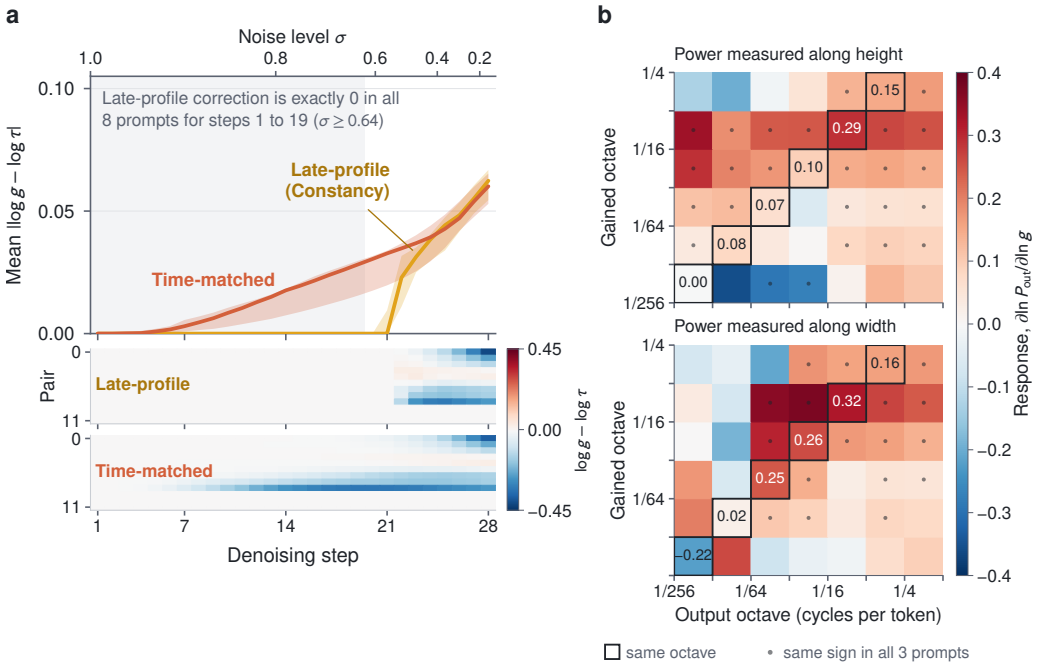


Figure F.1: The late-profile Constancy correction stays off for most of sampling while the time-matched target acts throughout, and attention-gain transfer is broad rather than band-selective. (a) Mean $|\ln g - \ln \tau|$ over spatial rotary pairs per step (median and interquartile band) on the eight dev prompts, seed 0, with both targets on the NTK base; below, per-pair correction maps. (b) Response of output-octave log power to a 1.5× logit gain on one octave, mean of 3 dev prompts, with power measured along height and width.

CPU study of the printed SEGA law. Before running any model we measured the printed allocation (Equation (2)) on forward-noised latents of the FLUX.2 autoencoder for 20 DIV2K crops. A fixed tilt over pairs explained 71% of the variance of the modulation, tilt plus timestep 91%, leaving at most 9.2% for content; the flatness gate ς was explained to $R^2 = 0.83$ by the timestep alone and to 0.78 by a generic $1/f$ field. On our held-out FLUX.1-Krea-dev runs the released gate ς is likewise explained to $R^2 = 0.89$ by the timestep. The channel mean that the printed sensor reads keeps 1.7% of the latent power. In a blur sweep at $\sigma = 0.15$ over 10 test crops, the printed law's contrast between

fine and coarse pairs changed by 12% between no blur and a blur of 2 tokens, while an oracle version of the constancy law changed 2.4-fold. These measurements motivated the constancy design; they use the printed law, another autoencoder and forward-noised rather than sampled latents, so we use them only as context for Section 5.2.

Probe protocol. The transfer probe of Figure F.1 reruns a dev prompt with the gain of one input octave multiplied by 1.5 on both spatial axes, with the same seed, and reports the finite-difference elasticity $\partial \log P_{\text{out}} / \partial \log g_{\text{in}}$ for six input and seven output octaves of the final latent and image. It used the NTK base with the log-N temperature, three dev prompts and one seed. The local-response experiment applied half-step perturbations at steps 6, 14 and 22 of two dev prompts and compared the observed change in band power with the prediction of a locally fitted linear response: the median error relative to predicting no change was 0.94 (range 0.71 to 1.35; 9 of 12 cases below 1). It tested a single octave, below the resolution of the token grid, so it does not cover all rotary bands. On the width axis of FLUX.1-Krea-dev, the mean diagonal elasticity of the transfer probe is 0.47 for one prompt but −0.03 and −0.04 for the other two, against a mean off-diagonal response of 0.09. Raising the gain of the fastest octave gives a width-axis elasticity of 0.11 on FLUX.1-Krea-dev and −3.64 in the same probe on FLUX.1-dev (one prompt; guidance and code revision also differ; Section 5.3).

Allocation controls on the dev set. A third dev campaign replaced the allocation of SEGA (released) by a uniform gain with the same root-mean-square value, or kept its shape and removed the rise in mean gain caused by the clamp. On the eight dev prompts the uniform control changed MUSIQ by −1.57 and the shape-only control by +0.39, while a rerun of SEGA (released) itself moved MUSIQ by +0.79. These differences are within run-to-run variation and were not taken to held-out prompts.

Diagnostics not reported. The project also measured run-to-run repeatability of the gain histories, the sensitivity of SEGA's sensor to channel-mean shifts at artificial states, attention-operator approximations and native-resolution semantic consistency. None bears on the claims of this paper, and we do not report them.

G Compute and timing

Table G.1 gives the sampling cost of the 1558 retained 4096^2 images, 79 GPU-hours in total (73 on GH200, 6 on GB200), with a median peak allocation of 40.8 GB. The second campaign's dev runs and the held-out SEGA + TM run used one GB200 node; every other run used GH200 nodes, and every forced-FlashAttention-2 timing at 4096^2 comes from the eight-prompt pilot. The GPU of each image was recovered from task and fleet logs, and its attention policy from the code revision pinned by its implementation hash, because the records carry a constant attention label (Section 5.5). The SDPA column of Table G.1 gives the requested backend policy (cuDNN first: cuDNN, then FlashAttention, then memory-efficient); the dispatched kernel was not logged. The SEGA family comprises released and paper-equation SEGA with their static ablations; the SEGA + TM family, SEGA + TM and its dev-only allocation controls; the Constancy family, Constancy and its dev sweeps, Constancy-TM, the temperature-only ablations, Alias-T and Alias-T-s. Node c619-102 ran 14% slower than the other GH200 nodes under both kernels (median power 665 W versus 836 W while generating), so it is listed separately; the forced-FA2 YaRN and DyPE pilots ran only there. The native rows run FLUX.1-Krea-dev without RoPE changes; the first image of each worker process carries a one-off warm-up (4.1 s with cuDNN first, 0.9 s with FA2), which widens the cuDNN-first IQR at 1024^2. The official DyPE script uses the default SDPA dispatch of diffusers. Timings exclude model loading (89 s per worker process) and PNG writing, except the single official DyPE run.

Table G.1: Sampling cost is set by the attention kernel and the GPU, not by the method. Seconds per image cover prompt encoding, 28 denoising steps and VAE decoding; IQR is the interquartile range; peak is `torch.cuda.max_memory_allocated`; GPU-h sums the per-image seconds; *n* counts images. SDPA policies, method families and the separately listed node are defined in the text.

SDPA	Methods	*n*	Nodes	Median (s)	IQR (s)	Peak (GB)	GPU-h
4096^2, *GH200 (10 nodes)*							
FA2, forced	NTK	8	1	289.8	289.8–289.9	40.8	0.6
FA2, forced	SEGA family	23	3	289.8	289.7–290.1	40.8	1.9
FA2, forced	Constancy family	16	2	290.4	289.9–290.5	40.8	1.3
cuDNN first	NTK	160	6	187.2	186.2–187.5	40.8	8.3
cuDNN first	DyPE	167	6	187.9	187.6–188.1	40.8	8.7
cuDNN first	SEGA family	197	9	187.8	187.5–188.2	40.8	10.3
cuDNN first	SEGA + TM family	24	2	189.4	184.9–189.6	40.8	1.3
cuDNN first	Constancy family	532	8	187.9	187.6–188.6	40.8	27.8
4096^2, *GH200 node c619-102 (listed separately)*							
FA2, forced	YaRN, DyPE	16	1	329.6	329.5–329.7	40.8	1.5
cuDNN first	all methods	195	1	214.6	214.0–215.3	40.8	11.6
4096^2, *GB200 (1 node, 4 GPUs)*							
cuDNN first	SEGA + TM family	204	1	93.3	92.5–94.3	40.8	5.3
cuDNN first	Constancy family	16	1	93.2	92.1–93.3	40.8	0.4
native 1024^2, *GH200, FLUX.1-Krea-dev*							
FA2, forced	no RoPE change	16	1	5.47	5.47–5.48		
cuDNN first	no RoPE change	8	2	4.94	4.93–5.93	36.3	
4096^2, *GH200, official DyPE script*							
default	DyPE (official code)	1	1	193.3			

H Reproducibility checklist

- **Numbers.** `paper/tables/make_tables.py` recomputes every held-out paired difference from the archived per-prompt scores, checks it against the registered analysis output (largest absolute difference below 10^{-15}), and writes every table and every number cited in the text. `compute_table.py` and `mechanism_numbers.py` do the same for timings and mechanism statistics. Scratch inputs are snapshotted next to the scripts.
- **Protocol.** Dev rows 0–7 and held-out rows 8–194 of the Aesthetic-4K evaluation captions; seed 0; seed-1 replication on rows 8–62; selection gate and its outcomes in Appendix D.2.
- **Generation.** Every image has a JSON sidecar with the full configuration, code revision, implementation hash, executed noise levels, per-step gains, wall time and peak memory.
- **Statistics.** Paired percentile bootstrap, 2,000 resamples, generator seed 0, per metric, no multiplicity correction.
- **Third-party code.** SEGA and DyPE are re-implemented from their released code at pinned revisions with numerical parity tests; DyPE's official code was also run end to end for the checkpoint diagnosis.
- **Not done.** No human study; one model family; one resolution; one seed for most arms; FID and KID too noisy at these sample sizes to rank methods.